

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxx} \\ M_{xxx} & -M_{xxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxx} & 0 & 0 \end{bmatrix}$$

$$M_{xxx} = \mathbb{M}_{4b}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxx} & M_{xxy} \\ 0 & 0 & 0 & 0 & -M_{xxx} & -M_{xxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzz} & M_{xxz} \\ -M_{xxx} & M_{xxx} & 0 & -M_{yzz} & 0 & 0 \\ -M_{xxy} & M_{xxy} & 0 & -M_{xxz} & 0 & 0 \end{bmatrix}$$

$$M_{xxx} = -2\mathbb{T}_{3b}^{(1)}$$

$$M_{xxy} = 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)}$$

$$M_{yzz} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzz} & 0 \\ -M_{yzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxx} & M_{xxy} \\ 0 & -M_{xxx} & M_{xxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzz} & 0 \\ M_{yzz} & 0 & 0 \\ M_{xxx} & 0 & 0 \end{bmatrix}$$

$$M_{xxx} = \mathbb{T}_{3b}^{(2)}$$

$$M_{xxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzz} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzzx} &= \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{yzxy} &= \mathbb{T}_{3b}^{(3)} \\ M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} & 0 & M_{xxxz} & M_{xxyx} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} & 0 & -M_{xxxz} & -M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & -M_{yzxz} & 0 & 0 & -M_{yzxz} & 0 & 0 \\ -\frac{M_{xxxy}}{2} - \frac{M_{xxxy}}{2} & \frac{M_{xxxy}}{2} + \frac{M_{xxxy}}{2} & 0 & -M_{xxzx} & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{xxzx} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} + \mathbb{T}_{3b}^{(2)} \\ M_{xxxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\ M_{xxxz} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} - \mathbb{T}_{3b}^{(2)} \\ M_{xxyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\ M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\ M_{yzxz} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\ M_{yzxy} &= -\mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} \\ M_{yzxz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzxz} &= -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{yzxy} &= \mathbb{T}_{3b}^{(3)} \\ M_{yzxz} &= \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -M_{xxxz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxxz} & 0 & 0 \\ 0 & 0 & -M_{xxxz} \\ 0 & -M_{xxxz} & 0 \end{bmatrix}$$

$$M_{xxxz} = \mathbb{M}_{4b}^{(1)}$$